

CENG381 Stochastic Processes

Martingales and the Optional Stopping Theorem - Question Set 1

Notation used in this question set:

Filtration F_n:	F_n represents all information available up to and including time n . Formally it is the sigma-algebra generated by $X_1, \dots, X_n$.
Martingale $M(n)$:	a stochastic process adapted to F_n such that $E[M(n)] < \infty$ and $E[M(n+1) F_n] = M(n)$ for all $n \geq 0$. The conditional expected future value always equals the present value (a 'fair game').
Supermartingale:	$E[M(n+1) F_n] \leq M(n)$ (expected value non-increasing -- 'unfair game').
Submartingale:	$E[M(n+1) F_n] \geq M(n)$ (expected value non-decreasing).
Stopping time T:	a random variable T taking values in $\{0, 1, 2, \dots, \infty\}$ such that the event $\{T = n\}$ depends only on F_n (i.e. you can tell whether to stop at time n using only information available by time n).
Optional Stopping Thm:	If $M(n)$ is a martingale and T is a stopping time satisfying either (i) T is bounded ($T \leq N$ for some fixed N), or (ii) $E[T] < \infty$ and $ M(n+1) - M(n) \leq C$ for some constant C , then $E[M(T)] = E[M(0)]$.

OST says: the expected value at a stopping time equals the initial value. You cannot gain an advantage over a fair game by clever stopping.

Q1 (30 points). Let $X_1, X_2, \dots$ be i.i.d. with $P(X_i = +1) = P(X_i = -1) = 1/2$ (symmetric random walk steps). Define $S(n) = X_1 + X_2 + \dots + X_n$ with $S(0) = 0$.

- Show that $S(n)$ is a martingale by verifying $E[S(n+1) | F_n] = S(n)$. Use the fact that X_{n+1} is independent of F_n and has $E[X_{n+1}] = 0$.
 - Define $M(n) = S(n)^2 - n$. Show $M(n)$ is also a martingale. (Hint: write $S(n+1)^2 = (S(n) + X_{n+1})^2$, expand, take conditional expectation, and use $E[X_{n+1}^2] = 1$ and $E[X_{n+1}] = 0$.)
 - Now suppose X_i has $E[X_i] = c > 0$ (a biased walk drifting upward). Without computing, state whether $S(n)$ is a martingale, supermartingale, or submartingale. What about $-S(n)$? Justify briefly.
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Q2 (35 points). A gambler plays a symmetric random walk starting at position k , with absorbing barriers at 0 (ruin) and N (goal). Let T = first time $S(n)$ reaches 0 or N . Assume $E[T] < \infty$ (it can be shown that T is always finite a.s.).

- Apply the Optional Stopping Theorem to the martingale $S(n)$ to find $h(k) = P(\text{reach } N \text{ before } 0 \mid S(0) = k)$. Use: $E[S(T)] = E[S(0)] = k$, and $S(T) = N$ with probability $h(k)$ or $S(T) = 0$ with probability $1-h(k)$.
 - Apply the Optional Stopping Theorem to the martingale $M(n) = S(n)^2 - n$ to find $E[T \mid S(0) = k]$, the expected number of steps to absorption. Use: $E[M(T)] = E[M(0)] = k^2$, and $S(T)^2 = N^2$ with prob $h(k)$ or $S(T)^2 = 0$ with prob $1-h(k)$. Express $E[T]$ in terms of k, N .
 - For $N = 4$ and $k = 1$, compute $h(1)$ and $E[T \mid S(0)=1]$. Interpret: starting one step from ruin and three steps from the goal, how likely is the gambler to succeed, and how long does the game last?
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Q3 (35 points). Consider the 'Stop When You're Ahead' strategy: a player bets on a fair coin repeatedly. Let $S(n)$ = total winnings after n bets. The player uses stopping time $T^* = \min\{n \geq 1 : S(n) > 0\}$, i.e. they stop as soon as they are ahead.

- Is T^* a valid stopping time? Explain why the event $\{T^* = n\}$ is determined by the outcomes $X_1, \dots, X_n$ only.
- What is $S(T^*)$ whenever $T^* < \infty$? (That is, what are the possible values of S at the moment of stopping?) Since steps are ± 1 , what is the only possible value of $S(n)$ at the first positive time?
- Despite $S(n)$ being a martingale and T^* being a stopping time, we have $E[S(T^*)] = 1 \neq 0 = S(0)$. Which condition of the OST fails? Show that $E[T^*] = \infty$ for the symmetric random walk (by arguing that a symmetric random walk is recurrent and the expected return time to any state is infinite).